E.G.S.PILLAYENGINEERINGCOLLEGE

(Autonomous)

NAGAPATTINAM– 611 002.

(Affiliated to Anna University, Chennai|Accredited by NAAC with 'A++'Grade
Accredited by NBA |Approved by AICTE, New Delhi)

REGULATIONS -R2023

B.E./B.Tech.–FIRST YEAR CURRICULUM

ELECTRONICS AND COMMUNICATION ENGINEERING

SEMESTER I									
COURSECODE	COURSE NAME	CATEGORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
Theory Courses									
2301IP101	Induction Program	-	0	0	0	0	0	0	0
2301MA102	Linear Algebra, Calculus and Partial Differentiation	BSC	3	1	0	4	40	60	100
2301PH101	Physics of Semiconductor and Optoelectronic Devices	BSC	3	0	0	3	40	60	100
2301CH103	Chemistry for Electronics Engineers	BSC	3	0	0	3	40	60	100
2301GEX03	Problem solving using C	ESC	2	0	4	4	50	50	100
2301ENX01	Professional English	HSMC	2	0	0	2	40	60	100
2301TA101	Heritage of Tamils / தமிழர் மரபு	HSMC	1	0	0	1	100	0	100
Laboratory Courses									
2301PHX51	Engineering Physics Laboratory	BSC	0	0	2	1	60	40	100
2301CHX51	Engineering Chemistry Laboratory	BSC	0	0	2	1	60	40	100
2301ENX51	Communication Skills Laboratory	HSMC	0	0	2	1	100	0	100
2301GEX52	Engineering Practices Laboratory	ESC	0	0	4	2	60	40	100
2301LS101	Life Skills –I	-	0	0	0	0	100	0	100
TOTAL			**14**	**1**	**14**	**22**	**690**	**410**	**1100**

2301MA102	LINEAR ALGEBRA,CALCULUS AND PARTIALDIFFERENTIATION (Common to ECE AND BME)	L	T	P	C
		3	1	0	4

PREREQUISITE:

1. Matrices
2. Differentiation
3. Integration.

COURSE OBJECTIVES:

1. To educate Matrix Algebra Technique and curvature theory.
2. To familiarize the students with differential calculus to apply in solving Modern Engineering Problems.
3. To acquaint the student with mathematical tools needed in evaluating multiple integrals and their applications.
4. To familiarize the student with functions of several variables.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Apply Orthogonal Transformation to determine the system of linear equations with matrices.
CO2:	Develop the evolutes and envelopes of given curves using radius and center of curvature.
CO3:	Apply the integration concepts to solve real life problems that arise in the field of engineering
CO4:	Calculate the nature of series using comparison, Ratio, Leibnitz tests.
CO5:	Examine the maxima/minima of multivariable function.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									
CO2	3	2	1									
CO3	3	2	1									
CO4	3	2	1									
CO5	3	2	1									

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	1		
CO2	1		
CO3	1		
CO4	1		
CO5	1		

COURSE CONTENTS:

MODULE I	**MATRICES**	**9 Hours**

Rank of a matrix - Consistency of the system of linear equations - linear dependence and independence of vectors. Eigen values and Eigen vectors of a matrix - Caley-Hamilton theorem and its applications - Reduction to diagonal form - Reduction of a quadratic form to canonical form - orthogonal transformation.

MODULE II	**DIFFERENTIAL CALCULUS**	**9 Hours**

Curvature in Cartesian co-ordinates – Centre and radius of curvature – Circle of curvature-Evolutes andinvolutes.

MODULE III	MULTIPLE INTEGRALS	9 Hours
Double and triple integrals - computation of surface areas and volumes- Change the order of Integration.		
MODULE IV	**SEQUENCE AND SERIES**	**9 Hours**
Sequences : Definition and Examples – series : Types and Convergence – Series of positive terms – Tests of Convergence: Comparison test, Integral test and D'Alembert's ratio test – Alternating series – Leibnitz's test – Series of positive and negative terms – Absolute and conditional convergence.		
MODULE V	**FUNCTIONS OF SEVERAL VARIABLES**	**9 Hours**
Partial differentiation - Total Differentiation – Euler's theorem and generalization. Maxima and minimaof functions of several variables (two and three variables) – Taylor's series for functions of two variables -Lagrange's method of Multipliers.		
		TOTAL:45 + 15 = 60 HOURS

REFERENCES:

1.Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2018.

2.R. K. Jain and S. R. K. Iyengar, "Advanced Engineering Mathematics", 5th ed, Narosa Publishing House, 2016.

3.G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson,Reprint, 2002.

4.Erwin kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley &Sons,2006.

5.Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11thReprint, 2010.

6.D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.

7.N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, LaxmiPublications, Reprint, 2008.

8.B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.

2301PH101	PHYSICS OF SEMICONDUCTOR AND OPTOELECTRONIC DEVICES	L	T	P	C
	(Common to ECE & BME)	3	0	0	3

PREREQUISITE:

1. Basic knowledge in physics

COURSE OBJECTIVES:

1.To instill knowledge on physics of semiconductors, determination of charge carriers and device applications

2.To make the students to understand the basics of dielectric materials, electrical properties of materials including free electron theory, applications of quantum mechanics

3.To establish a sound grasp of knowledge on different magnetic & optical properties of materials

4.To make the students to understand the basics of optoelectronic devices, optical displays and applications

5.To inculcate an idea of significance of nano structures, quantum confinement and ensuing nano device applications.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Understand clearly of semiconductor physics and functioning of semiconductor devices
CO2:	Apply basics of dielectric materials, gain knowledge on the electrical properties of materials and their applications

CO3:	Understand the magnetic, optical properties of materials
CO4:	Demonstrate a strong knowledge in optoelectronic devices and working principles of various optical devices
CO5:	Apply the importance of nanotechnology and nanodevices

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									
CO2	3	2	1									
CO3	3	2	1									
CO4	3	2	1									
CO5	3	2	1									

COs Vs PSOs MAPPING

Cos	PSO1	PSO2	PSO3
CO1	1		
CO2	-		
CO3	-		
CO4	1		
CO5	1		

COURSE CONTENTS:

MODULE I	**SEMICONDUCTORS AND TRANSPORT PHYSICS**	**9 Hours**

Intrinsic Semiconductors – Energy band diagram – direct and indirect band gap semiconductors – Carrier concentration in intrinsic semiconductors – extrinsic semiconductors - Carrier concentration in N-type & P-type semiconductors – Variation of carrier concentration with temperature – Carrier transport in Semiconductors: Drift, mobility and diffusion – Hall effect and devices – Ohmic contacts – Schottky diode.

MODULE II	**DIELECTRIC MATERIALS AND ELECTRICAL PROPERTIES**	**9 Hours**

Polarization mechanisms: electronic, ionic, orientational, interfacial and total polarization – frequency dependence – local field and Causius-Mossetti equation – dielectric constant and dielectric loss. Classical free electron theory - Expression for electrical conductivity – Thermal conductivity, expression - Quantum free electron theory - Electron in periodic potential – Energy bands in solids

MODULE III	**MAGNETIC & OPTICAL PROPERTIES OF MATERIALS**	**9 Hours**

Magnetic materials: Dia, para and ferromagnetic effects – paramagnetism in the conduction electrons in metals – exchange interaction and ferromagnetism – quantum interference devices – GMR devices.
Classification of optical materials – Optical processes in semiconductors: optical absorption and emission, charge injection and recombination, optical absorption, loss and gain. Optical processes in quantum wells

MODULE IV	**OPTOELECTRONIC DEVICES**	**9 Hours**

Optoelectronic devices: light detectors and solar cells – light emitting diode – laser diode – optical processes in organic semiconductor devices –excitonic state – Electro-optics and nonlinear optics: Modulators and switching devices – plasmonics

MODULE V	**NANO DEVICES**	**9 Hours**

Density of states for solids - Significance between Fermi energy and volume of the material – Quantum confinement – Quantum structures – Density of states for quantum wells, wires and dots – Band gap of nanomaterials –Tunneling – Single electron phenomena – Single electron Transistor-Carbon nanotubes: Properties and applications - Spintronic devices and applications – Optics in quantum structures – quantum well laser.

TOTAL: 45 HOURS

REFERENCES:

1.S.O. Kasap. Principles of Electronic Materials and Devices, McGraw Hill Education (Indian Edition), 2020.

2. R.F.Pierret. Semiconductor Device Fundamentals. Pearson (Indian Edition), 2006.

3. G.W.Hanson. Fundamentals of Nanoelectronics. Pearson Education (Indian Edition), 2009.

4. Jasprit Singh, Semiconductor Optoelectronics: Physics and Technology, McGraw- Hill Education (Indian Edition), 2019.

5. Charles Kittel, Introduction to Solid State Physics, Wiley India Edition, 2019

6. https://archive.nptel.ac.in/courses/108/108/108108122/
7. https://onlinecourses.nptel.ac.in/noc20_ph24/preview

2301CH103	CHEMISTRY FOR ELECTRONICS ENGINEERS	L	T	P	C
		3	0	0	3

PREREQUISITE:

- Basic knowledge of science up to higher secondary level

COURSE OBJECTIVES:

- To make the students conversant with boiler feed water requirements, related problems and water treatment techniques
- To impart technological aspects of applied chemistry

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Describe the amount of ion present in the water sample.
CO2:	Determine the pH of the solutions.
CO3:	Illustrate corrosion using electrochemical principles.
CO4:	Describe the construction and working of storage devices.
CO5:	Summarize nano technology and polymer materials.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										
CO2	2	1										
CO3	2	1										
CO4	2	1										
CO5	2	1										

COs Vs PSOs MAPPING

COs	PSO1	PSO2
CO1		
CO2		
CO3		
CO4	1	
CO5	1	1

COURSE CONTENTS:

MODULE I	WATER TECHNOLOGY	9 Hours

Hardness of water – types – expression of hardness – units – estimation of hardness of water by EDTA -Alkalinity- boiler troubles (scale and sludge) – treatment of boiler feed water – Internal treatment (phosphate, colloidal, sodium aluminate and calgon conditioning) external treatment – Ion exchange process, zeolite process – desalination of brackish water- Reverse Osmosis.

MODULE II	ELECTROCHEMISTRY	9 Hours

Cell terminology-Electrochemical cells- Electrolytic cells- Cell reactions- Daniel cell-Difference between electrolytic cells and electrochemical cells. Reversible cells and irreversible cells -types- EMF and its applications - Nernst equation (derivation and problems).Single electrode potential - Hydrogen electrode - Calomel electrode - Glass electrode - pH measurement using glass electrode.

MODULE III	CORROSION	9 Hours

Corrosion – principles of corrosion – Pilling – Bed worth rule – principles of electrochemical corrosion – difference between chemical and electrochemical corrosion – galvanic corrosion – differential aeration corrosion – factors influencing corrosion – corrosion control – cathodic protection – sacrificial anodic method.		
MODULE IV	**DEVICE FABRICATION**	**9 Hours**
Batteries – introduction, Representation of a battery, Types- Alkaline battery, Lead acid, Nickel–Cadmium and Lithium ion batteries–advantages and disadvantages. Fuel Cells – Basic Structure- Hydrogen Fuel Cell. Renewable energy resources, photovoltaic cell -application.		
MODULE V	**NANOTECHNOLOGY AND POLYMER MATERIALS**	**9 Hours**
Nanotechnology: Basics - distinction between molecules, nanoparticles and bulk materials; size-dependent properties. Nano particles: nano cluster, nano rod, nanotube(CNT) and nanowire. Synthetic methods: chemical vapour deposition, laser ablation; synthesis of metal oxide nano particles Polymer -functionality –degree of polymerisation- molecular weight determination (weight average and number average)-Thermoplastic & Thermo setting- Nanoparticles embedded polymer composites, conducting Polymers		
		TOTAL: 45 HOURS

REFERENCES:

1.Sashi Chawla, A Text book of Engineering Chemistry, Dhanpat Rai Publishing Co., Pvt. Ltd., Educational and Technical Publishers, New Delhi, 3rd Edition, 2015.

2.S. S. Dara, *A Text book of Engineering Chemistry*, S. Chand & Co Ltd., New Delhi, 20th Edition, 2013.

3.P.C. Jain and Monica Jain, A Textbook of Engineering Chemistry, DhanpatRai publications, New Delhi, 16th edition, 2015.

4.O.V. Roussak and H.D. Gesser, *Applied Chemistry-A Text Book for Engineers and Technologists*, Springer Science Business Media, New York, 2nd Edition, 2013.

5.Alain Nouailhat, "An Introduction to Nanoscience and Nanotechnology", John Wiley, ISBN:978-1848210073

https://onlinecourses.nptel.ac.in/noc23_bt31/preview

https://onlinecourses.nptel.ac.in/noc23_me46/preview

2301GEX03	**PROBLEM SOLVING USING C (E-BOX)**	L	T	P	C
	(Theory cum Laboratory course)	2	0	4	4

PREREQUISITE:

Need some Mathematical Knowledge

COURSE OBJECTIVES:

- To understand the constructs of C Language.
- To develop C programmes using arrays and strings
- To develop modular applications in C using functions and pointers
- To develop applications in C using structures and union

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Demonstrate the knowledge about the techniques used to solve problems in computing.
CO2:	Build programmes using C constructs.
CO3:	Design and implement applications using arrays and strings
CO4:	Develop and implement modular applications in C using functions and pointers.
CO5:	Develop programmes and applications in C using structures, union.

COURSE CONTENTS:

MODULE I	PROGRAMMING BASICS	6 Hours

Data Types – Constants – Keywords – Expressions – Type of Errors - Structure of C programme – Preprocessor directives - Compilation process, Execution of source code. Operators and operator's precedence – I/O statements

MODULE II	CONTROL STRUCTURES	6 Hours

Sequence statements – Selection statements – Looping statements – Solve Numerical / Logical problems.

MODULE III	ARRAYS AND STRINGS	6 Hours

Introduction to Arrays: Declaration, Initialization – One dimensional array – Two dimensional array
String operations: length, compare, concatenate, copy, upper case, lower case.

MODULE IV	POINTERS AND FUNCTIONS	6 Hours

Pointers – Pointer operators – Pointer arithmetic – Arrays and pointers – Parameter passing: Pass by value, Pass by reference.
Function prototypes - function definition, function call – Recursion: Binary search using recursive functions

MODULE V	STRUCTURES AND UNION	6 Hours

Structure - Nested structures – Pointer and Structures – Array of structures – Dynamic memory allocation – Union - Storage classes and Visibility.

TOTAL: 30 HOURS

List of Experiments:

Prepare programmes in C to implement basic concepts in C language.	6 Hours
Produce C programmes to implement decision making and branching statements.	6 Hours
Use the concept of looping to implement C programmes.	6 Hours
Employ the concept of arrays to develop C programmes.	6 Hours
Experiment the concepts of strings using C.	6 Hours
Develop C programmes to perform code reusability using function.	6 Hours
Model programmes in C to implement pointers.	6 Hours
Build C programme to implement structures.	6 Hours
Build C programme to implement Union.	6 Hours
Mini Project: Structures, Functions & Pointers.	6 Hours

TOTAL: 60 HOURS

Hardware/software requirement

1. Desktop Systems 60 Nos
2. C Compiler

TOTAL: 90 HOURS

COs Vs POs / PSOs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1						2				3	1	
CO2	2	1						2				3	1	
CO3	3	2	3					2				3	1	
CO4	3	2	3					2				3	1	
CO5	3	2	3					2				3	1	

REFERENCES:

1.Yashwant Kanetkar, Let us C, 17th Edition, BPB Publications, 2020.

2.ReemaThareja, "Programming in C", Oxford University Press, Second Edition, 2016.

3.Kernighan, B.W and Ritchie,D.M, “The C Programming language”, Second Edition, Pearson Education, 2015.

4.Byron S. Gottfried, “Schaum’s Outline of Theory and Problems of Programming with C”, McGraw-Hill Education, 1996.

https://onlinecourses.nptel.ac.in/noc23_cs53/course (Link for NPTEL/SWAYAM/MOOC Courses)

https://cse02-iiith.vlabs.ac.in/exp (Link for virtual Lab)

www.skillrack.com (Link for modern tool usage)

2301ENX01	PROFESSIONAL ENGLISH	L	T	P	C
	Common to B.E /B.Tech Programme (CIVIL,BME,CSE,ECE,EEE,IT,MECH AND AIDS)	2	0	0	2

PREREQUISITE:

1. Basic English Knowledge

COURSE OBJECTIVES:

- To improve the communicative competence of learners.
- To learn using of basic grammatical structures in suitable contexts.
- To acquire lexical competence and use them appropriately in a sentence and understand their meaning in a text.
- To help learners in using the language effectively in professional contexts.
- To use the language efficiently in expressing their opinions.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Use appropriate words in a professional context
CO2:	Gain understanding of basic grammatical structures and use them in right context.
CO3:	Read and interpret information presented in tables, charts and other graphic forms
CO4:	Write definitions, descriptions, narrations and essays on various topics
CO5:	Speak fluently and accurately in formal and informal communicative contexts.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1										3		
CO2										3		
CO3										3		
CO4										3		
CO5										3		

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	-	-	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-

COURSE CONTENTS:

MODULE I	FUNDAMENTALS OF COMMUNICATION	9 Hours
Reading - Reading brochures (technical context)/ user manuals/, telephone messages / social media messages relevant to technical contexts and emails. Writing - Professional emails etiquette, emails / letters (seeking permission for Industrial visit& Complain letter) Grammar - Present Tense (simple and progressive); Question types: Why/ Yes or No/ and Tags. Vocabulary - Word forms (prefixes& suffixes); Synonyms and Antonyms, Punctuation.		
MODULE II	**NARRATION AND SUMMATION**	**9 Hours**
Reading - Reading longer technical texts (Reading biographies/ travelogues/ newspaper reports/ travel & technical blogs). Writing - Paragraph writing Short Report on an event (field trip etc.), emails / letters (Writing responses to complaints). Grammar –Past tense (simple); Subject-Verb Agreement. Vocabulary – Preposition, Prepositional Phrases& Phrasal verbs.		
MODULE III	**DESCRIPTION OF A PROCESS / PRODUCT**	**9 Hours**
Reading – Reading advertisements, gadget reviews. Writing – instructions, Checklists, Report Writing (Accident Report & Survey Report (IV)). Grammar – Present & Past Perfect Tenses, Voices (Active ,Passive & Impersonal Passive Voice); Vocabulary – Collocations, Homonyms; and Homophones,		
MODULE IV	**CLASSIFICATION AND RECOMMENDATIONS**	**9 Hours**
Reading – Newspaper articles; Journal reports –and Non Verbal Communication (tables, pie charts etc,).; Writing- Job / Internship application – Cover letter &Resume ,recommendations. Grammar – Articles, Adjectives of Comparison, If conditional sentences Vocabulary – Conjunctions, discourse markers (connectives & sequence words)		
MODULE V	**EXPRESSION**	**9 Hours**
Reading – Company profiles, standard operating procedure (SOP)/ an excerpt of interview with professionals. Writing – Essay Writing (Descriptive or narrative), Grammar – Future Tenses, Numerical adjectives, Relative Clauses. Vocabulary - Cause & Effect Expressions – Content vs Function words.		
		TOTAL: 45 HOURS

REFERENCES:

1. Technical Communication – Principles And Practices By Meenakshi Raman & Sangeeta Sharma, Oxford Univ. Press, 2016, New Delhi.
2. A Course Book On Technical English By Lakshminarayanan, Scitech Publications (India) Pvt. Ltd.
3. English For Technical Communication (With CD) By AyshaViswamohan, Mcgraw Hill Education, ISBN : 0070264244.
4. Effective Communication Skill, Kulbhusan Kumar, RS Salaria, Khanna Publishing House.
5. Learning to Communicate – Dr. V. Chellammal, Allied Publishing House, New Delhi,2003.
6. Raman. Meenakshi, Sharma. Sangeeta (2019). Professional English. Oxford university press. New Delhi
7. New Delhi. 2. Improve Your Writing ed. V.N. Arora and Laxmi Chandra, Oxford Univ. Press, 2001, NewDelhi.

2301TA101	தமிழர்மரபு/ HERITAGE OF TAMILS	L	T	P	C
		1	0	0	1

PREREQUISITE:

Cultural and Heritage of Tamil People. And about Sangam Literature, Religion, Art & Architecture during ancient Period. And also about Classical Tamil Literarture,Epigraphy, Inscriptions, Non-Tamil source

COURSE OBJECTIVES:

- To preserve and expose the cultural heritage of our ancestors and teachers
- To evoke surprise and admiration by creating accessibility to their work, products, and minds.
- To promote the practice of science and other cultural activities.

COURSE OUTCOMES:

At the end of this course, students will be able to

CO1: To use Learning and understanding the development of modern literature.

CO2:	To acquire the kinds of modern literature and the development of modern literature.
CO3:	To learn the Tamil Drama Literature, its origin, development;authors Understand the difference between drama of historic period and nowadays. Impact of modern theories in theatre art
CO4:	To develop the medieval literature and to know the impact of religious literature through the ages
CO5:	To speak the human values and rights in Tamil literature. Learn to respect the people who live with them Respect the people without the basis Gender, Race and economic status.

Cos Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1										3		
CO2										3		
CO3										3		
CO4										3		
CO5										3		

COURSECONTENTS:

MODULE I	LANGUAGE AND LITERATURE	**3 Hours**

Language Families in India - Dravidian Languages – Tamil as a Classical Language – Classical Literature in Tamil–Secular Nature of SangamLiterature– Distributive Justice in Sangam Literature - Management Principles in Thirukural - Tamil Epics and Impact of Buddhism & Jainism in TamilLand-Bakthi Literature Azhwars and Nayanmars-Forms of minor Poetry-Development of Modern literature in Tamil-Contribution of Bharathiyar and Bharathidhasan.

அலகு I மொழி மற்றும் இலக்கியம்: 3

இந்திய மொழிக் குடும்பங்கள் – திராவிட மொழிகள் – தமிழ் ஒரு செம்மொழி – தமிழ் செவ்விலக்கியங்கள் - சங்க இலக்கியத்தின் சமயச் சார்பற்ற தன்மை – சங்க இலக்கியத்தில் பகிர்தல் அறம் – திருக்குறளில் மேலாண்மைக் கருத்துக்கள் – தமிழ்க் காப்பியங்கள், தமிழகத்தில் சமண பௌத்த சமயங்களின் தாக்கம் - பக்தி இலக்கியம், ஆழ்வார்கள் மற்றும் நாயன்மார்கள் – சிற்றிலக்கியங்கள் – தமிழில் நவீன இலக்கியத்தின் வளர்ச்சி – தமிழ் இலக்கிய வளர்ச்சியில் பாரதியார் மற்றும் பாரதிதாசன் ஆகியோரின் பங்களிப்பு.

MODULE II	**HERITAGE-ROCK ART PAINTINGS TO MODERN ART–SCULPTURE**	**3 Hours**

Hero stone to modern sculpture - Bronze icons - Tribes and their handicrafts - Art of temple car making - Massive Terracotta sculptures, Village deities, Thiruvalluvar Statue at Kanyakumari,Making of musical instruments - Mridhangam, Parai, Veenai, Yazh and Nadhaswaram - Role of Temples in Social and Economic Life of Tamils.

அலகு II மரபு – பாறை ஓவியங்கள் முதல் நவீன ஓவியங்கள் வரை – சிற்பக் கலை: 3

நடுகல் முதல் நவீன சிற்பங்கள் வரை – ஐம்பொன் சிலைகள்– பழங்குடியினர் மற்றும் அவர்கள் தயாரிக்கும் கைவினைப் பொருட்கள், பொம்மைகள் – தேர் செய்யும் கலை – சுடுமண் சிற்பங்கள் – நாட்டுப்புறத் தெய்வங்கள் – குமரிமுனையில் திருவள்ளுவர் சிலை – இசைக் கருவிகள் – மிருதங்கம், பறை, வீணை, யாழ், நாதஸ்வரம் – தமிழர்களின் சமூக பொருளாதார வாழ்வில் கோவில்களின் பங்கு.

அலகு III நாட்டுப்புறக் கலைகள் மற்றும் வீர விளையாட்டுகள்: 3

தெருக்கூத்து, கரகாட்டம், வில்லுப்பாட்டு, கணியான் கூத்து, ஒயிலாட்டம், தோல்பாவைக் கூத்து, சிலம்பாட்டம், வளரி, புலியாட்டம், தமிழர்களின் விளையாட்டுகள்.

MODULE IV	**THINAI CONCEPT OF TAMILS**	**3 Hours**

Flora and Fauna of Tamils & Ahamand Puram Concept from Tholkappiyam and Sangam Literature
Aram Concept of Tamils - Education and Literacy during Sangam Age - Ancient Cities and Ports of Sangam Age-Export and Importduring Sangam Age -Overseas Conquest of Cholas.

அலகு IV தமிழர்களின் திணைக் கோட்பாடுகள்: 3

தமிழகத்தின் தாவரங்களும், விலங்குகளும் – தொல்காப்பியம் மற்றும் சங்க இலக்கியத்தில் அகம் மற்றும் புறக் கோட்பாடுகள் – தமிழர்கள் போற்றிய அறக்கோட்பாடு – சங்ககாலத்தில் தமிழகத்தில் எழுத்தறிவும், கல்வியும் – சங்ககால நகரங்களும் துறை முகங்களும் – சங்ககாலத்தில் ஏற்றுமதி மற்றும் இறக்குமதி – கடல்கடந்த நாடுகளில் சோழர்களின் வெற்றி.

MODULEV	**CONTRIBUTION OF TAMILS TO INDIAN NATIONAL MOVEMENT AND INDIANCULTURE**	**3 Hours**

Contribution of Tamils to Indian Freedom Struggle-The Cultural Influence of Tamils over the otherparts of India – Self-Respect Movement – Role of Siddha Medicine in Indigenous Systems of Medicine– Inscriptions & Manuscripts–Print History of Tamil Books.

அலகு V இந்திய தேசிய இயக்கம் மற்றும் இந்திய பண்பாட்டிற்குத் தமிழர்களின் பங்களிப்பு: 3

இந்திய விடுதலைப்போரில் தமிழர்களின் பங்கு – இந்தியாவின் பிறப்பகுதிகளில் தமிழ்ப் பண்பாட்டின் தாக்கம் – சுயமரியாதை இயக்கம் – இந்திய மருத்துவத்தில், சித்த மருத்துவத்தின் பங்கு – கல்வெட்டுகள், கையெழுத்துப்படிகள் - தமிழ்ப் புத்தகங்களின் அச்சு வரலாறு.

TOTAL:15HOURS

REFERENCES:

1. தமிழகவரல ொறு– மக்களும் பண் லபொடும்– மக.மக. பிள் மள (தவளியீடு: தமிழ்ல ொடுலபொடநூ ்மற்றும்க ்வியிய ்பணிகள்கழகம்).
2. கணினித்தமிழ் – முமனவரஇ் . சு ்தரம். (விகடன் பிரசுரம்).
3. கீழடி– மவமக திக்கமரயி ல ங்கலகொ கரல ொகரிகம் (தலதொ ்லிய ்Fமற தவளியீடு)
4. தலபொரும – ஆற்றங் கமரல ொகரிகம். (தலதொ ்லிய ்Fமறதவளியீடு)
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6. Social Life of the Tamils - The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu) (Published by: International Institute of Tamil Studies).
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies)
9. Keeladi - 'Sangam City C ivilization on the banks of river Vaigai' (Jointly Published by: Department ofArchaeology& Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Publishedby: The Author)
11. Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Bookand Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) – Reference Book.

Laboratories

2301PHX51	**ENGINEERING PHYSICS LABORATORY**	**L**	**T**	**P**	**C**
	(Common to all branches)	**0**	**0**	**2**	**1**

PREREQUISITE:

1. Basic knowledge in physics

COURSE OBJECTIVES:

1. To learn the proper use of various kinds of physics laboratory equipment
2. To learn how data can be collected, presented and interpreted in a clear and concise manner.
3. To learn problem solving skills related to physics principles and interpretation of experimental data.
4. To determine error in experimental measurements and techniques used to minimize such error.
5. To make the student an active participant in each part of all lab exercises

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Apply the theoretical concepts of physics in procedures and techniques in performing the experiments
CO2:	Apply and demonstrate, thermal conductivity, electrical properties of metals and semiconductors, elastic properties of materials and oscillations through experiential learning
CO3:	Demonstrate the use of monochromatic light, lasers in optical fiber communication and quantum mechanics towards specific engineering applications
CO4:	Use different measuring devices/ meters to record the data with precision and apply the mathematical concepts/equations to obtain quantitative results
CO5:	Develop basic communication skills through working in groups in performing the laboratory experiments and by interpreting the results

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	1	1	2	2			
CO2	3	3	2	2	1	1	1	2	2			
CO3	3	3	2	2	1	1	1	2	2			
CO4	3	3	2	2	1	1	1	2	2			
CO5	3	3	2	2	1	1	1	2	2			

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1			
CO2			
CO3			
CO4			
CO5			

LIST OF EXPERIMENTS (Any 7 experiments to be performed)

1. Torsional pendulum - Determination of rigidity modulus of wire and moment of inertia of regular and irregular objects.
2. Simple harmonic oscillations of cantilever.
3. Non-uniform bending - Determination of Young's modulus
4. Uniform bending – Determination of Young's modulus
5. Laser- Determination of the wavelength of the laser using grating
6. Air wedge - Determination of thickness of a thin sheet/wire
7. a) Optical fibre -Determination of Numerical Aperture and acceptance angle b) Compact disc- Determination of width of the groove using laser.
8. Acoustic grating- Determination of velocity of ultrasonic waves in liquids.

9. Ultrasonic interferometer – determination of the velocity of sound and compressibility of liquids
10. Determination of Band gap of a semiconductor.
11. Poiseuille's method for finding viscosity of a liquid
12. Lee's Disc-Thermal conductivity of bad conductor
13. Spectrometer-determination of wavelength using grating

2301CHX51	ENGINEERING CHEMISTRY LABORATORY	L	T	P	C
	COMMON TO ALL BRANCHES	0	0	2	1

PREREQUISITE:

- Basic knowledge of science up to higher secondary level

COURSE OBJECTIVES:

- To make the students conversant with boiler feed water requirements, related problems and water treatment techniques
- To impart technological aspects of applied chemistry

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Estimate the amount of ion present in the water sample.
CO2:	Determine the pH of the solutions.
CO3:	Estimate the corrosion behavior of metals.
CO4:	Determine the acid content using electrochemical principles.
CO5:	Determine the molecular weight of the polymer.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			1			1	1			
CO2	3	2			1			1	1			
CO3	3	2			1			1	1			
CO4	3	2			1			1	1			
CO5	3	2			1			1	1			

COs Vs PSOs MAPPING

COs	PSO1	PSO2
CO1		
CO2		
CO3	1	
CO4		
CO5		

LIST OF EXPERIMENTS

1. Determination of total, temporary & permanent hardness of water by EDTA method
2. Comparison of alkalinities of the given water samples
3. Estimation of iron content of the given solution using potentiometer
4. Corrosion experiment – weight loss method
5. Conductometric titration of strong acid Vs strong Base
6. Determination of molecular weight of a polymer by viscometry method
7. Determination of percentage of copper in alloy
8. Determination of ferrous iron by Spectrophotometry method

9. Estimation of calcium present in cement.
10. Determination of strength of given hydrochloric acid using pH meter
11. Estimation of sodium ion present in water by flame photometer.
12. Estimation of dissolved oxygen in a water sample/sewage by Winklers method.
13. Synthesis of metal oxide nanoparticles by chemical method.

REFERENCES:

1. Experimental organic chemistry, Daniel R. Palleros, John Wiley & Sons, Inc., New Yor (2001)
2. Engineering Chemistry", Jain & Jain, 15th edition, Dhanpat Rai Publishing company, New Delhi
3. Vogel"s Textbook of practical organic chemistry, Furniss B.S. Hannaford A.J, Smith P.W.G and Tatchel A.R LBS Singapore (1994)
4. LBS Singapore (1994). Kolthoff I.M., Sandell E.B. et al Mcmillan, Madras 1980

2301ENX51	COMMUNICATION SKILLS LABORATORY	L	T	C
	Common to B.E /B.Tech Programme (CIVIL,BME,CSE,ECE,EEE,IT,MECH and AI&DS)	0	0	1

PREREQUISITE:

1. Basic English Knowledge

COURSE OBJECTIVES:

CO1	To facilitate computer-aided multi-media instruction enabling individualized and independent language learning
CO2	To bring about a consistent accent and intelligibility in their pronunciation of English by providing an opportunity for practice in speaking.
CO3	To prepare them to use communicative language and participate in different types of speaking environments.
CO4	To expose the Students to participate in group discussions, debates with ease.
CO5	To enable the students become strong in LSRW skills.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Improve their listening, reading, speaking and writing skills.
CO2:	Develop their communication competency.
CO3:	Use language effectively in professional contexts.
CO4:	Develop the ability to face campus interviews.
CO5:	Use language efficiently in expressing their opinions

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1										3		
CO2										3		
CO3										3		
CO4										3		
CO5										3		

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	-	-	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-

CO5	-	-	-	

COURSE CONTENTS:

MODULE I	LISTENING	6 Hours

Listening for general information-specific details Audio / video (formal & informal).
Listening IELTS/TOFEL/ TED Talks and educational videos.
Listening to podcasts, anecdotes / stories / event narration; documentaries and interviews with celebrities.
Listening - Listen to product and process descriptions; and advertisements about products.
Listening – Listening to debates/ discussions; different viewpoints on an issue; and panel discussions.

MODULE II	SPEAKING	6 Hours

Self-Introduction - Role play exercises based on workplace contexts- Group discussion (Discussing advantages and disadvantages/ purposes and reasons)- Discussing progress toward goals- Discussing past events in life- Making telephone calls (politeness strategies- making polite requests, making polite offers, replying to polite requests and offers) Interpreting (Picture, locations in workplaces).

MODULE III	READING	6 Hours

Reading– Intensive Reading -Comprehending general and technical articles -Cloze reading - Inductive reading- Short narrative and descriptions from newspapers – Skimming and scanning-reading and interpretation-Critical reading Interpreting and transferring graphical information- Sequencing of sentences..

MODULE IV	WRITING	6 Hours

Writing- Precise writing –Summarizing- Interpreting visual texts (pie chart, bar chart, picture, advertisements etc., - Proposal writing (launching new units or department in a institution or industry & to get loan from bank) -Report writing (accident, progress, project, survey, Industrial visit)- Job application- Resume.

MODULE V	PERSONALITY DEVELOPMENT	6 Hours

Introduction to life skills -emotional intelligence (visualizing and experiencing purpose)-Self-awareness - Time management-Stress management -Leadership- teamwork & dealing with ambiguity--interview planning- Mock Interviews-– Self-Concept.
Organizational etiquette.

TOTAL: 30 HOURS

REFERENCES:

1.Effective Communication Skill, Kulbhusan Kumar, RS Salaria, Khanna Publishing House.
2.Learning to Communicate – Dr. V. Chellammal, Allied Publishing House, New Delhi,2003.
3.New Delhi. 2. Improve Your Writing ed. V.N. Arora and Laxmi Chandra, Oxford Univ. Press, 2001, NewDelhi.
4.Developing Communication Skills by Krishna Mohan, Meera Bannerji- Macmillan India Ltd. 1990, Delhi.
5.Business Correspondence and Report Writing by Prof. R.C. Sharma & Krishna Mohan, Tata McGraw Hill & Co. Ltd., 2001, New Delhi.
https://swayam.gov.in/explorer?searchText=english (Link for NPTEL/SWAYAM/MOOC Courses)
https://ieltsonlinetests.com (Link for modern tool usage)

2301GEX52	ENGINEERING PRACTICES LABORATORY	L	T	P	C
		0	0	4	2

PREREQUISITE: NIL

COURSE OBJECTIVES:

1. To provide hands on training for fabrication of components using sheet metal and welding equipment / tools.
2. To develop skill for using carpentry and fitting tools to make simple components and metal joints.
3. To provide training for making simple house hold pipe line connections using suitable tools.
4. To develop the skill to make / operate/utilize the simple engineering components.

COURSE OUTCOMES:

At the end of this course, Students will be able to,

CO1: Fabricate simple components using sheet metal using suitable tools.

CO2: Prepare simple components using suitable fitting tools.

CO3: Fabricate simple components using welding equipments.

CO4: Make simple components / joints using carpentry power tools.

CO5: Make simple house hold pipe line connections using suitable tools.

COs Vs POs & PSOs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2								2			1	-	-	-
CO2	2	1							2			1	-	-	-
CO3	2	1			1				2			1	-	-	-
CO4	2	1			1				2			1	-	-	-
CO5	2								2			1	-	-	-

LIST OF EXPERIMENTS	
1. Forming of simple object in sheet metal using suitable tools.(Example: Dust Pan, Rectangular tray and Cone making)	**6 Hours**
2. Prepare V (or) Half round (or) Square (or) Dovetail joint from the given mild Steel flat.	**5 Hours**
3. Fabrication of a simple component using thin and thick plates using arc welding. (Example: Butt , Lap and T - Joints)	**6 Hours**
4. Making a simple component using carpentry power tools.(Example: Cross Lap, T-Lap, Dove tail joints and Electrical switch box / Tool box / Letter box)	**6 Hours**
5. Construct a household pipe line connections using pipes, Tee joint, four way joint, elbow, union, bend, Gate valve and Taps.	**5 Hours**
6. Study of gas welding equipment and its demonstration.	**2 Hours**
	Total : 30 Hours

References:

1. S. Gowri &T.Jeyapoovan, “Engineering Practices Lab Manual” 5th Edition,Vikas Publishing.
2. Dr. V. Ramesh Babu,”Engineering Practices Laboratory Manual” Revised Edition 2019-20, VRB Publishers Pvt. Ltd.